

Exercise12

CPS I — Exercise Sheet 12

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Exercise 1 :Done

Exercise 2: Fairness

F1 : For (a) to (f) all false.

F2

a:Yes, it is not enabled.

b:Yes, δ is enabled and executed

c.Yes, we have execution. $s_0 \xrightarrow{\beta} s_2 \xrightarrow{\delta} \left(s_1 \xrightarrow{\delta} s_2 \xrightarrow{\delta} \right)^\omega$ It satisfies weak fairness for δ

d.No, δ is infinite enabled since s_1 , but only execute only once.

e.Yes, the same in (c). We have execution. $s_0 \xrightarrow{\beta} s_2 \xrightarrow{\delta} \left(s_1 \xrightarrow{\delta} s_2 \xrightarrow{\delta} \right)^\omega$ It satisfies weak fairness for δ

f.Yes, the same in (c). We have execution. $s_0 \xrightarrow{\beta} s_2 \xrightarrow{\delta} \left(s_1 \xrightarrow{\delta} s_2 \xrightarrow{\delta} \right)^\omega$ It satisfies weak fairness for δ

F3:

a:Yes, not enabled.

b:No, infinite enabled but not executed.

c:Yes, we have execution $s_0 \xrightarrow{\alpha} (s_1 \xrightarrow{\gamma})^\omega$

d:Yes, infinite enabled and executed.

e:Yes, same in (d)

f:Yes, same in (d)

F4:

a:Yes

b:No, η is infinite enabled but not executed.

c:No, $F4 = F2 \cap F3$ Even though F2 and F3 both satisfy (c), but in different executions.

d:No, δ infinite enabled but not execute.

e:Yes, execution $s_0 \xrightarrow{\beta} s_2 \xrightarrow{\eta} (s_3 \xrightarrow{\eta})^\omega$

f: Yes, same in (e)

Exercise 3: Fairness

1. weak fairness for "enter"

- Sufficiency. "Enter" is always enabled in "w", we don't want it to do self loop on "w", so "Enter" must infinite execute.
- Weakest. Weak Fairness is the weakest. Without weak fairness for "enter", it will do self loop on "w".

2. Strong fairness for "enter"

- Sufficiency. We don't want the case $(w_1 \rightarrow w_2 \rightarrow)^\omega$. If "Enter" is infinite enabled, it must infinite executed.
- Weakest. Weak fairness for "enter" is not possible. Because in the case $(w_1 \rightarrow w_2 \rightarrow)^\omega$ "Enter" is infinite enabled, but not always enabled.

3. Not possible to give fairness assumption on action enter to ensure non-starvation?